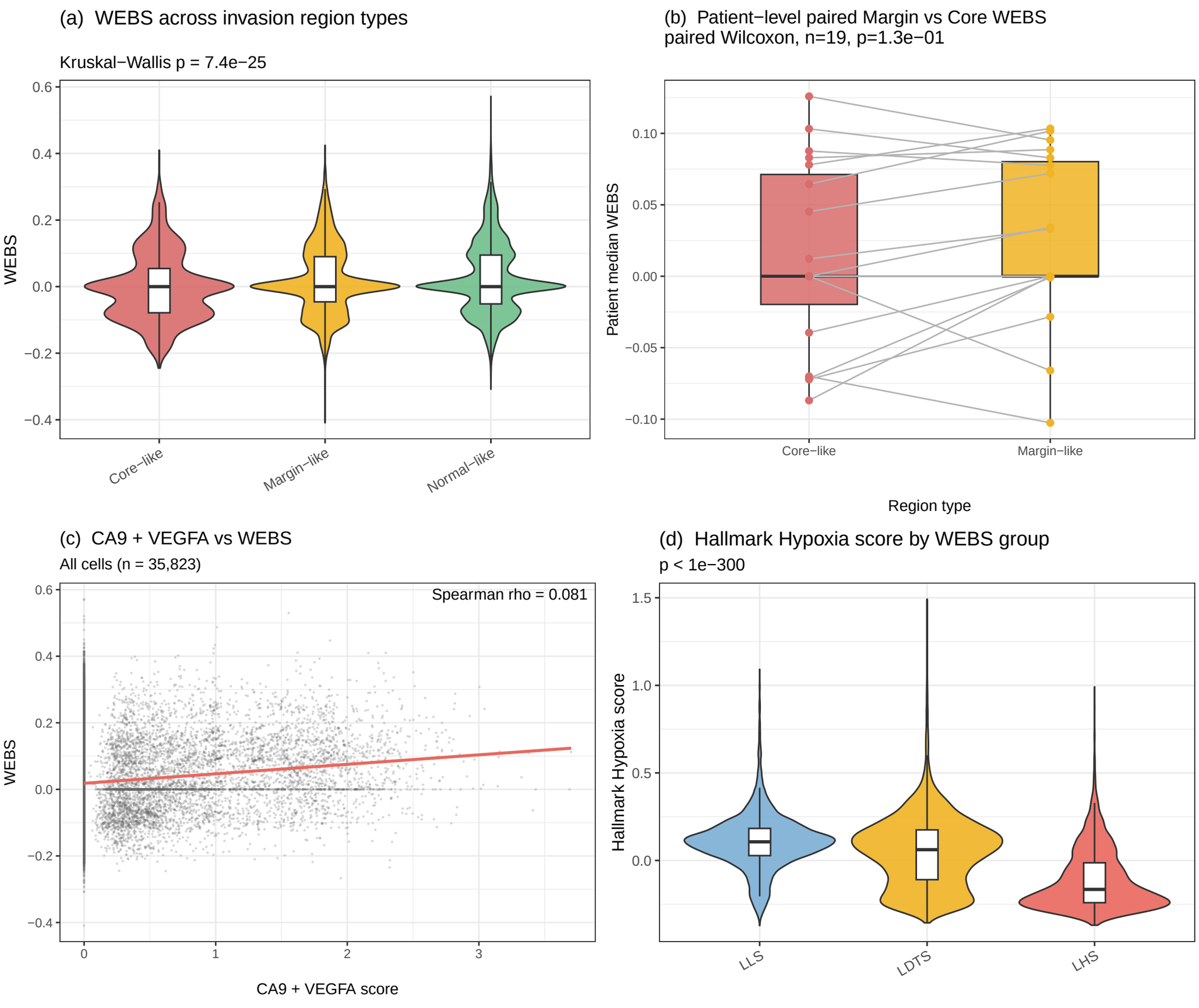

(a) WEBS across invasion region types
Kruskal−Wallis p = 7.4e−25
WEBS
0.6
0.4
0.2
0.0
−0.2
−0.4
Core−like
Margin−like
Normal−like
(b) Patient−level paired Margin vs Core WEBS
paired Wilcoxon, n=19, p=1.3e−01
Patient median WEBS
0.10
0.05
0.00
−0.05
−0.10
Core−like
Margin−like
Region type
(c) CA9 + VEGFA vs WEBS
All cells (n = 35,823)
Spearman rho = 0.081
WEBS
0.6
0.4
0.2
0.0
−0.2
−0.4
0
1
2
3
CA9 + VEGFA score
(d) Hallmark Hypoxia score by WEBS group
p < 1e−300
Hallmark Hypoxia score
1.5
1.0
0.5
0.0
LLS
LDTS
LHS